



Healthcare Data Cleaning & Exploratory Data Analysis

Project by: Vikas Prajapati | **Date:** April 22, 2026

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn)

Why Data Cleaning Matters



Raw Data Reality

Data in raw form is often messy and inconsistent, requiring cleaning before analysis.



EDA Purpose

Exploratory Data Analysis helps understand patterns and extract meaningful insights.



Project Focus

Cleaning and analyzing healthcare data using Python for better decision-making.



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Occqations

Data

- Clean and preprocess raw healthcare dataset
- Handle missing and inconsistent data
- Perform comprehensive EDA
- Extract meaningful insights from the dataset

Dataset Overview

The dataset contains healthcare-related records including patient and hospital information.



Patient Demographics

Age and gender information for patient profiling



Medical Details

Medical condition, admission date, and medication tracking



Financial Data

Billing amount and insurance provider information

Data Cleaning Process



Remove Duplicates

Eliminated duplicate records from the dataset



Correct Names

Fixed name column formatting and spelling



Fix Data Types

Converted incorrect data types for consistency



Standardize Names

Standardized column names across the dataset

Exploratory Data Analysis

EDA Approach

- Analyzed distribution of data across variables
- Identified trends and patterns in healthcare records
- Compared different variables for correlations
- Used visualization for better understanding



Visualizations & Analysis

1

Bar Charts

Patient count by gender and medical condition distribution by gender

2

Heat Maps

Medical condition versus test results and medication frequency analysis

3

Histogram

Billing amount distribution showing frequency patterns



Key Insights

50-65

Age Group

Majority of patients belong to this age range

27K

Gender Balance

Equal patient count for Male and Female

Most Used Medication per Condition

Arthritis

Aspirin (1,901)

Asthma

Paracetamol (1,870)

Cancer

Lipitor (1,904)

Diabetes

Lipitor (1,875)

Hypertension

Ibuprofen (1,874)

Obesity

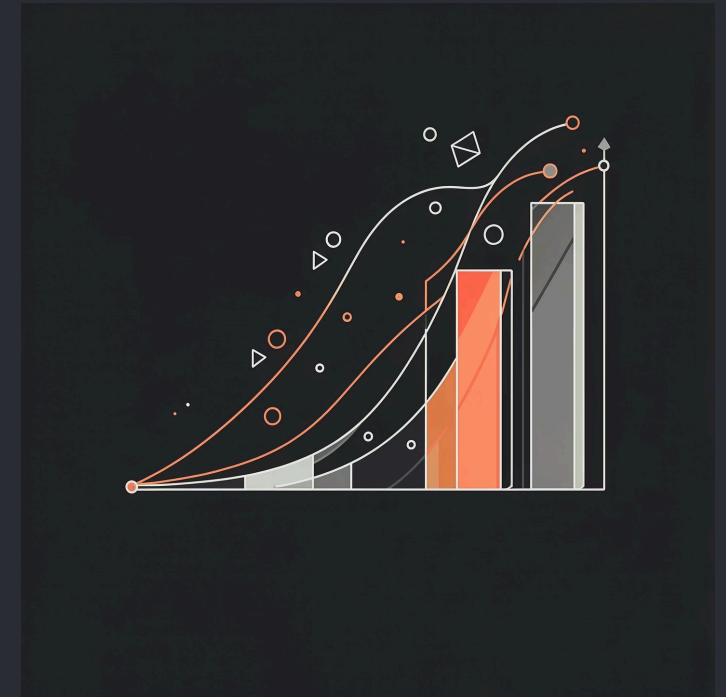
Penicillin (1,872)

Analysis Findings

Billing Amount Distribution

The histogram shows a roughly bell-shaped distribution centered around \$25,000-\$30,000, with the highest frequency occurring in this range.

Pattern: Unimodal and slightly right-skewed distribution



100%

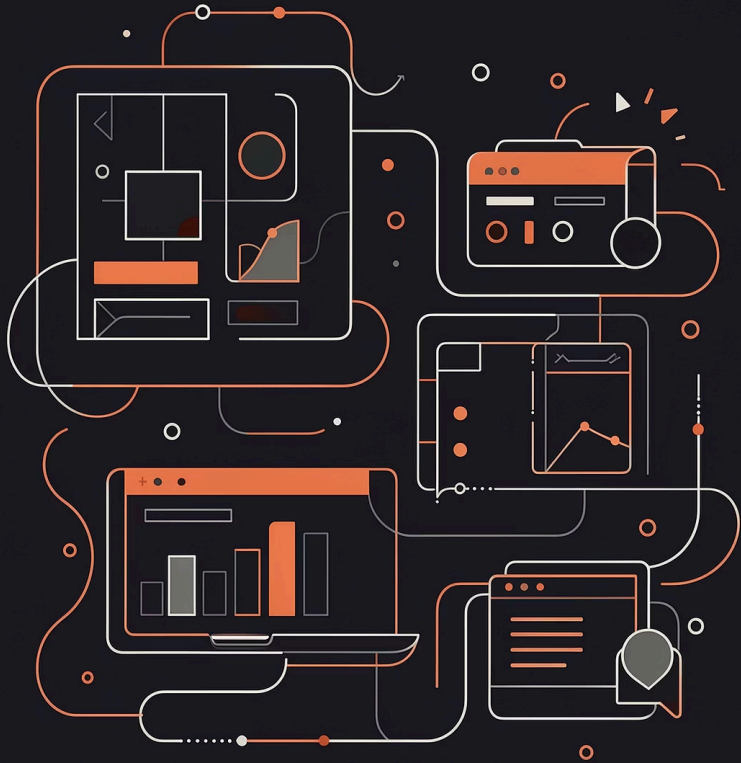
Equal Gender Distribution

Both Male and Female patients at 27,000 each

100%

Uniform Condition Distribution

All six conditions at 4,800 patients per gender



Conclusion

01

Data cleaning is essential before analysis

02

EDA helps uncover useful insights

03

Python is effective for data analysis tasks

04

Clean data leads to better decision-making